Answer Key (Tutorial - 4)

Q.1 $y[n] = \begin{cases} n-6 & 7 \le n \le 11 \\ 6 & 12 \le n \le 18 \\ 24-n & 19 \le n \le 23 \\ 0 & \text{else} \end{cases}$

Q.2 $y(t) = \begin{cases} t+3 & -2 < t \le -1 \\ t+4 & -1 < t \le 0 \\ 2-2t & 0 < t \le 1 \\ 0 & \text{else} \end{cases}$

Q.3 (a) $y[n] = g[n-2]$

(b) $y[n] = g[n-4]$

(c) S/S is not LTI

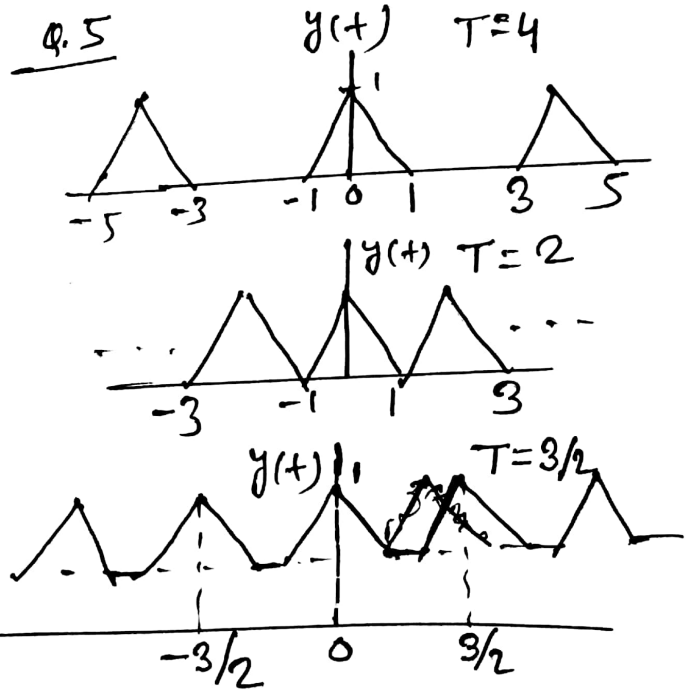
(d) $y[n] = \sum_{k=0}^{\infty} g[n-2k]$

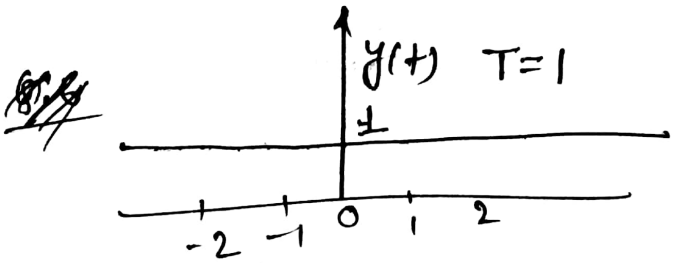
$= \begin{cases} 1 & n = 0, 1 \\ 2 & n > 1 \\ 0 & \text{else} \end{cases}$

Q.4 (a) $A = \frac{1}{5}$

(b) $g[n] = \delta[n] - \frac{1}{5}\delta[n-1]$

Q.5





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Q.11

(a) True

(b) False

(c) False

(d) True

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Q.6

(a) C + stable

(b) NC + stable

(c) C + unstable

(d) C + stable

Q.7

(a) C + stable

(b) NC + unstable

(c) NC + stable

Q.8

(a) $h(t) = e^{-(t-2)} u(t-2)$

(b) $y(t) = \begin{cases} 0 & t<1 \\ 1-e^{-(t-1)} & 1<t<4 \\ e^{-(t-4)}(1-e^{-3}) & t>4 \end{cases}$

Q.9

(a) $\omega_0 = 2\pi$, $y(0)=0$

(b) not unique

for any $\omega_0 = 2k\pi$, $k \in I$ and $k \neq 0$

will give $y(0)=0$

Q.10

(c) $h(t) = \omega_0 \cos(\omega_0 t) + 5\sin(\omega_0 t)$

(e) $y(t) = \left(e^{-3t} - 2e^{-2t} + 1\right)u(t) + \left[\frac{1}{4}\left(e^{t} - e^{-3t}\right) - \frac{2}{3}\left(e^{t} - e^{-2t}\right) - e^{t} - 1\right]u(t)$

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